

Geometry of Splats

Splats functions are *mean-field two layer neural networks!*

Some unique phenomenology:

- Highly sparse, localized activation patterns.
- Neurons are smoothed particles. Can teleport, prune, etc.

Geometry of Splats

Chizat Bach 2018: conditional global optimization for 2-layer ReLU/Sigmoid nets via WGF, for,

$$\mathcal{F}(\mu) = R \left(\int \Phi d\mu \right) + \int V d\mu \quad \mu \in \mathcal{P}_2(\Omega), \Omega \subset \mathbb{R}^d$$

Here: $\Phi(v, W, b) = v(Wx + b)_+$, risk $R(\cdot)$, regularizer $V(\cdot)$